ID N1

Function and characteristics of an ultrasound distance sensor (Maxbotix XL-MaxSonar-EZ)

Learning objectives

After completing this laboratory, you should be able to:

1. Experimentally characterize an ultrasonic distance sensor
2. Determine and evaluate its output / distance characteristic
3. Analyze sensor output using a multimeter and an oscilloscope
4. Acquire sensor data using a microcontroller ADC
5. Analyze measurement noise statistically using Python
6. Interpret the received ultrasonic waveform and determine the speed of sound from the measured time-of-flight

Before coming to the lab, you must be able to

1. Explain the principle and relevant parameters of Time-of-Flight measurements with ultrasound
2. Distinguish the terms accuracy, precision, and resolution of a sensor
3. Draw a characteristic curve to describe the expected behavior of the MB1360
4. Draw a circuit diagram including a microcontroller and the electrical connections of the MB1360
5. Explain the relation of an analog sensor output voltage and an ADC-value, describe ADC resolution and quantization
6. Measure a DC voltage with a multimeter
7. Explain the purpose of using an oscilloscope for signals not constant over time
8. Calibrate a probe
9. Compile and upload an Arduino program
10. Read serial data from an Arduino in Python
11. Plot a time-series and a histogram, calculate mean, median and standard deviation in Python

Equipment

- Arduino Nano
- Ultrasound sensor Maxbotix XL-MaxSonar- EZ MB1360 (https://maxbotix.com/products/mb1360?srsltid=AU7gw4Ucc4ymd0lkkEBD38meNEG7lsveq_5g84jwnNqh_H-9YvY8g8lr, https://cdn.shopify.com/s/files/1/0550/8091/0899/files/PD11840l-XL-MaxSonar-EZ_Datasheet-_website_1.pdf?v=1679586742)
- Breadboard
- USB-mini cable, connector cables
- Ruler (distance reference)
- Free space with a flat object as test object for distance measurement (wall, cardboard box, …)
- Lab PC or Laptop, Arduino-IDE, Python-IDE (Anaconda, Spyder)
- Multimeter, Oscilloscope, Probes

Total time: approx. 1.5 h

Setup

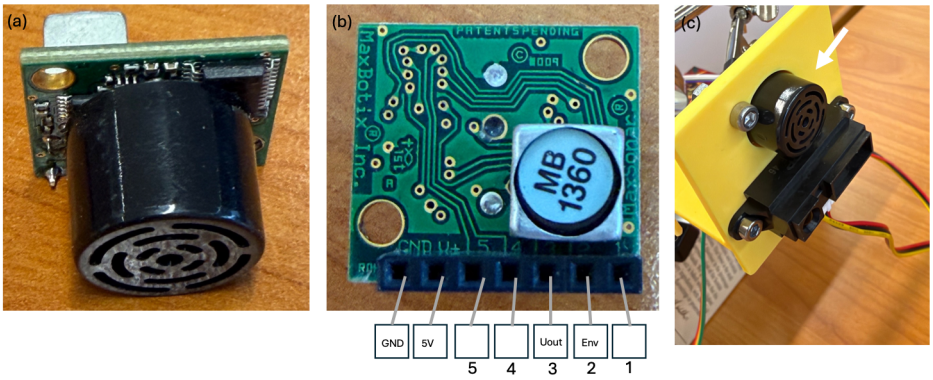


Figure 1: (a) Maxbotix Ultrasound Sensor (b) pinout (c) mechanical setup

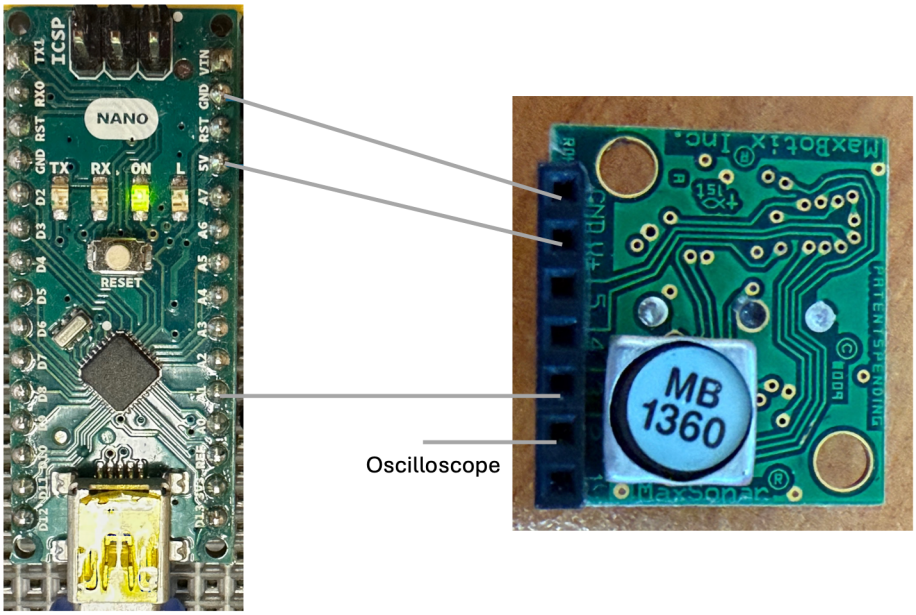


Figure 2: Connections

The Arduino provides the 5V supply voltage and GND reference for the sensor. The sensor output Uout is measured with a multimeter (exercise 1), an oscilloscope (exercise 2), and with the ADC of the microcontroller (exercise 3). Furthermore, the Env-Output – providing the waveform of the received echo - is to be measured with the oscilloscope (exercise 4).

Exercise 1 –Sensor Setup, Characteristic Curve, Multimeter Measurements

- Connect the sensor lines as indicated (Supply Voltage, GND, Analog Sensor Output Uout)
- Put a square object in front of the sensor and vary the distance between the sensor front and the object for a range from 0 cm to 200 cm. Measure the sensor output Uout.
- Select at least 10 measurement distances between 0 cm and 200 cm. Choose the distances so that important features of the sensor characteristic can be identified. Do not simply use equally spaced distances. Justify your choice of measurement points.
- Measure the sensor output Uout as a function of distance
- Compare your result with the sensor specification.

Deliverables

- Photos of your setup with annotations for the connections and the full view on sensor, object, and the measurement range
- Table and characteristic curve Uout(distance) measured with the multimeter
- Assessment of the design of experiment and the results:
    1. Why did you select the measurement distances?

2. Within which distance range can the sensor output be used reliably for distance measurement? Justify your answer.
3. How well does your characteristic curve fit to the curve specified in the sensor documentation?
4. What is the maximum deviation between the expected and the measured output voltage observed? What is the resulting error in cm?

Exercise 2 – Characteristic Curve, Oscilloscope Measurements

- Compensate the oscilloscope probes
- Add the oscilloscope to your setup to measure the behavior of the sensor output Uout over time
- Add the automatic measurement of minimum, mean, and maximum to the oscilloscope trace
- Observe the temporal behavior of Uout. Based on the observed waveform, discuss whether a sampling rate of approximately 10 kHz is sufficient to read the sensor output correctly. What type of problems would you expect when measuring sensor output with a A/D-converter running at about 10 kHz sampling rate?
- Use the oscilloscope math functions to measure min/mean/max of the voltage output Uout of the sensor for at least 10 different distances in the range 0 cm to 200 cm. Choose the same values for the distance as in exercise 1
- Compare your results with the sensor specification and with your result of exercise 1

Deliverables

- Photos of your setup with annotations for the connections and the full view on sensor, object, and the measurement range
- Screenshots of the oscilloscope measurements Uout(t) for at least five different distances and your observation
- Table and characteristic curves Uout(distance) measured with the oscilloscope for minimum, mean, maximum values of Uout
- Assessment of the results:
    1. How do your results fit to the results obtained in exercise 1?
    2. In which range does the sensor provide a suitable output signal?
    3. How well does your characteristic curve fit to the curve specified in the sensor documentation?
    4. What is the maximum deviation from the specified curve observed?
    5. Which additional information does the oscilloscope measurement provide? What does that mean for measuring the output with an ADC?

Exercise 3 – Sensor and Arduino Setup, ADC converter

- Setup the Arduino and the ultrasound distance sensor on the breadboard
- Connect three pins of the sensor V+, GND, Uout to the respective pins on the Arduino (check in the software and adjust values for pins if necessary)
- Load the Arduino code into the IDE, set all parameters for the board, and compile and transfer the code to the microcontroller
- Use “Werkzeuge/Serieller Monitor“ or “Werkzeuge/Serieller Plotter“ to check the output of the sensor.
- What happens, if you intentionally mismatch the baud-rate of Arduino and the serial monitor?
- What is the maximum baud-rate to allow stable communication?

Deliverables

- Screenshot of the serial monitor for matched and for mismatched parameters
- What is the maximum possible baud-rate for stable communication. Please justify your answer.
- How does the measured voltage Uout get into the PC: Draw a block diagram and data flow diagram for your setup

Exercise 4 – Statistical measurement and data analysis

- Download the python code and open it in Anaconda/Spyder

- Identify the serial port used by the Arduino and set the port and all transmission parameters correctly
- Start the program and check the values printed on the screen. You may have to install the libraries „pyserial“, „numpy“, and „matplotlib“
- Do not filter, smooth, reject or otherwise modify the raw ADC-values. Any data processing must be explicitly documented.

Deliverables

- Time series, histogram, and statistical indicators (mean, median, min, max, standard deviation) for a measurement distance of 50 cm with at least 100 continuously measured, raw ADC-values under constant conditions. Are 100 values enough to obtain a sufficient statistical characterization?
- Description: What do you observe on the time series of the sensor output? What are the consequences? What countermeasure would you propose?
- Characteristic curve of the sensor based on the mean ADC-value of 100 measurements as the output (use the same 10 distances as in exercise 1)

Exercise 5 – Measurement of the waveform received

- Connect the oscilloscope to the Env-pin
- Setup all relevant parameters of the oscilloscope to obtain a regularly updated and stable measurement
- Change the distance of your obstacle to the sensor and observe changes on the oscilloscope
- Find the echo received from the object.

Deliverables

- At least three annotated screenshots of the oscilloscope for different measured distances
- Mark the part of the waveform created by your obstacle.
- Determine the time interval between transmission and reception of the echo. Using the known distance between sensor and object and the measured round-trip time, calculate the speed of sound.